

Daniel Iván Parra Verde

ML Engineer • Production ML • forecasting • RAG/agents • since 2023 (+3 years)
Portfolio: danivpv.com | GitHub: danivpv | LinkedIn: danivpv | HuggingFace: danivpv
Tel. +52 844 783 6082 | danivpv@outlook.com | Saltillo, México

EDUCATION

UNAM-CFATA

2021 | BSc. TECHNOLOGY

GRONINGEN UNIVERSITY (G)

2019 | EXCHANGE (AI/CS)

Netherlands; Full Scholarship

UNAM-IIMAS

2022 | MSc. CANDIDATE IN MATH

SKILLS

AI and ML Libraries:

PyTorch • LangChain • LlamaIndex •
Hugging Face • scikit-learn • Pandas •
NumPy • Docling • Gemini • OpenAI •
Anthropic • OpenCV

Frameworks:

FastAPI • Temporal • Spark • Next.js •
SQLAlchemy • Django • Kafka

Cloud Services:

AWS EMR • AWS S3 • AWS SageMaker
• AWS Lambda • AWS CloudWatch •
AWS API Gateway • AWS ECS

Databases:

PostgreSQL • MongoDB • Redis •
Qdrant • SQLite

Programming Languages:

Python • SQL • Bash • TypeScript •
Terraform • Julia • Wolfram

Software Tools:

Docker • Git • GitHub Actions • Sentry
• Postman • $\LaTeX$ • Notion

Spoken Languages

English (+C1) • German (+A2) • Spanish

COURSEWORK

Bachelor

Deep Learning in **Python**(G)
Relational Databases in **SQL** (G)
Artificial Intelligence in **Python** (G)
Algorithms and Data Structures in **C** (G)
Linear Algebra

Master

Statistical Inference
Theoretic Machine Learning
Information Theory
Statistical Foundations of Privacy
Numerical Analysis **Python**, **Julia**
Real and Functional Analysis

WORK EXPERIENCE

SENIOR DATA SCIENTIST

Arkham Technologies | April 2025 – Present | Remote

- Engineered enterprise Machine Learning pipelines; scaled 1K models on **AWS EMR** via **Spark** and event-driven architecture (**Kafka**, **Temporal**).
- Deployed AI models into production and automated AI agent monitoring (LLM-as-Judge) via **FastAPI** microservices and **PostgreSQL**.
- Led **MLOps** & **CI/CD** of **Docker** images for prod ML platform via **GitHub Actions** and **Terraform**. 10× faster builds vs baseline.
- Engineered production AI integrations for cross-provider LLMs powering autonomous AI agents (100+/week users).
- Debugged **FastAPI/Kafka** ML and AI microservices via **CloudWatch** and **Sentry**.

AI MENTOR & DEVELOPER

LIDR | November 2024 – April 2025 | Remote

- Deliver slides and live-coding sessions on AI-assisted software development.

DATA SCIENTIST

Kuona | November 2023 – June 2024 | Remote

- Tuned **PyTorch RNNs** for enterprise sales forecasting, turning predictions into actionable decision intelligence for retail revenue management (12% MAPE).
- Modeled competitors' pricing data, reducing forecasting MAPE by 40% and enabling price-demand elasticity analysis for business decisions via **Pandas**.
- Designed QLoRA and sales-weighted loss functions for critical revenue KPIs.

DATA SCIENTIST JR.

Entropia AI | February 2023 – November 2023 | Remote

- Built 8 **ETL** pipelines (**Python**, **bs4**, **Airflow**) ingesting 2K docs with 100M+ tokens from official Mexican government sites into **MongoDB** for legal partners.
- Deployed **RAG** AI solutions into production; raised custom QA LLM-as-Judge accuracy from 8% to 72% on legal corpora using **LlamaIndex** and **MongoDB**.

SOFTWARE DEVELOPER (CONTRACT)

Wolfram Research | February 2021 – February 2022 | Remote

- Sysadmin. Fullstack dev of portal for "Wolfram Summer Camp" (+40 users).
- Published paper "Turing Patterns in Networks" in Complex Systems Journal.

PROJECTS

ARXIV DOMAIN EXPERT LLM

GitHub Repository

- Developed custom fine-tuning datasets in **Hugging Face**, fine-tuned Llama-3.2-3B on **AWS SageMaker** and deployed with vector database **Qdrant**.
- Automated experiment tracking using **ZenML**, **CometML** and **MongoDB**.

MASTER THESIS: NEURAL MODEL IDENTIFICATION WITH EKF

Ph.D. Esteban A. Hernandez-Vargas | Systems Medicine of Infectious Diseases

- Derived robust EKF weight update rule for RNNs, maintaining stability up to 20% signal-to-noise ratio in viral time-series modeling of SARS-Cov-2, HIV, etc...

"CONVOCATORIA JÓVENES TALENTOS 2018" CONCYTEC

Ph.D. Rafael Chávez Moreno | Unit of High Technology Juriquilla, Querétaro

- Secured \$89,000 MXN funding; led Rover CV/path planning and presentations.